

Event Ops Standards (all games)

The shared operational spine every TorqWars event runs on — applied to TorqFall, TorqSiege, and every future game. Covers Yash's "all games" asks: game technicalities, track/setup + dependency-free options, crew, full-day/atmosphere, and the "what am I missing" master checklist. Game-specific rules live in each rulebook; this is what's common.

1. Game-technicalities framework (spec every game against this)

A standard checklist so no game ships with a hidden failure mode. For each game, answer all 8:

1. **Radio / frequency** — many 2.4GHz radios in one hall = the #1 sleeper risk. Plan: hobby 2.4GHz auto-hops but a dense pack can still glitch → cap simultaneous cars, stagger bind/power-on, keep a "radio marshal," test max-pack binding in rehearsal. (See §5 — biggest missed item.)
2. **Power / run-time** — battery type, minutes per charge, swap cadence, how many battery sets per car to keep it always-on. Drives the pit size.
3. **Scoring & timing** — manual marshal call (default, fastest) vs stopwatch vs sensor gate vs transponder lap-timing. Pick the *lowest-tech method that's fair enough*; content-first tolerates a marshal's call.
4. **Marshal-to-car ratio** — how many staff to run + reset one heat safely.
5. **Reset time** — seconds between heats/rounds. The throughput number.
6. **Failure modes** — dead car mid-heat, flipped car, jammed obstacle, wall not collapsing (siege) → each needs a scripted marshal response.
7. **Safety envelope** — speed cap, soft-obstacle rule, crowd barrier line, LiPo handling.
8. **Content rig** — which cameras (overhead / finish-line / FPV / roving) capture *this* game's money shot.

2. Track / setup — own it, make it dependency-free



Arena render — v5 concept still, all four game zones around a central walkway

Philosophy: build-once, reuse, van-packable, and 2 sourcing options per critical item so no single venue fixture, supplier, or material can block an event ("free of dependability").

Surface (redundancy ladder):

- **Primary:** roll-out **astroturf** (grippy, safe, on-brand, reusable). ~3000 sq ft in sections.
- **Option B:** interlocking **EVA foam mats / tiles** (cushioned, modular, cheap, also our barrier material).
- **Option C:** **rubberised event flooring** (rented from the venue if they have it).
- **Fallback:** **bare polished floor + gaffer-tape lanes** (zero cost, less grip/look — last resort).

Barriers (redundancy ladder — soft, full-length, spectator-safe):

- Pool-noodle-on-stake rail · EVA foam block wall · inflatable tube barriers · hay bales (themed) · low plywood wall **faced with foam**. Keep ≥ 2 buildable from stock so a supplier gap never stops us.

Obstacle/structure kit: modular, labelled, stored in bins, packs in a van. (*Pointers updated 2026-09-06:*) the TorqFall obstacle-kit BOM lives in [TorqFall — What The Event Needs](#) §3; the TorqSiege castle/wall stock in [TorqSiege — What The Event Needs](#) §1 — not in Build & Fabrication Plan v1.

Power & light: venue mains + **a generator/battery fallback**; LED start/finish gates + uplights run off portable power so a venue with weak power can't gate us. Charging pit needs its own dedicated circuit (see §5).

Setup/teardown target: kit sets up in ≤ 4 hours by the crew, tears down in ≤ 2 . Rehearse the build once before event #1.

3. Crew (roster for a 100–300-person event)

Lean but complete — ~9–12 people. Roles (one person can double on a small event):

- **Event Lead** ×1 — owns the run-of-show, makes the calls.
- **Marshals** ×2–3 — run heats, reset obstacles, make rulings (the throughput engine).
- **Pit crew** ×1–2 — car repair, battery swap/charging, spares. *The unsung critical role* — a good pit = the event never stops for a dead car.
- **MC / Commentator** ×1 — **essential, not optional**. Turns "cars driving" into a *show*; hypes wipe-outs, runs the bracket, keeps energy. The single highest-leverage crew hire for a content-first event.
- **Filming** ×2 — one locked overhead, one roving/handheld. Content is the product; staff it like it.
- **Photographer** ×1 (can be a filming double) — stills for socials.
- **Queue / ticketing / registration** ×1–2 — check-in, waivers, heat assignment, data capture.
- **Safety marshal** ×1 — crowd line, LiPo/first-aid, "HOLD" authority.
- **Build-station host** ×1 — TorqSiege only (runs the build phase).
- **Sound / DJ** ×1 — see §4.

4. Full-day event — is the game solid alone, or do we add layers?

Verdict: the game is a solid CORE, but a full day needs an atmosphere layer — not a concert. A single game for hours risks *spectator* fatigue (players rotate through, but the room's energy has to be sustained). Right dose for a first event:

- **YES — MC/commentator** (§3): the cheapest, biggest energy multiplier. Non-negotiable.
- **YES — DJ / curated playlist + PA**: cheap, transforms the room, drives the "party" feel. A DJ for peak hours; a playlist otherwise.
- **YES — big-screen leaderboard/bracket + live camera feed**: makes it a *show*, pulls spectators in, and doubles as content.
- **YES — a day arc**: qualifying → bracket → a **Grand Final showpiece** at day's end gives the day a climax and a reason to stay (and the best reel).
- **YES — food/drink partner** (stall or co-located F&B): extends dwell, adds revenue, adds vibe. Partner, don't run it yourself.
- **MAYBE — a chill/spectator/photo zone** (branded backdrop, seating): low cost, boosts content + dwell.
- **NO (for event #1) — a full concert / big-name act**: expensive, splits focus, unproven ROI. Revisit once the format is proven and a flagship is wanted.
- **Content-first reminder**: the **camera is a guest too** — schedule the day around *capturing* peak moments (a set "final" everyone films), not just running heats.

5. "What are we missing" — master checklist (the sleepers)

Things easy to forget that can sink an event. Owner + status to be filled per event:

- **2.4GHz interference at scale** — TEST max-pack binding in rehearsal; it's the top technical sleeper.

- **Battery / charging logistics** — a fleet needs a charging pit, spare battery sets, rotation, and a dedicated power circuit. Plan minutes-of-runtime vs heats-per-hour.
- **LiPo fire safety** — LiPo bags, a sand bucket / fire extinguisher, no charging unattended. (Or spec NiMH for the fleet to sidestep most of this — decide in [Car Fleet, Scale & Protection](#).)
- **Insurance & liability** — public-liability cover + **signed waivers** for drivers (esp. minors → guardian consent).
- **Permits** — police permission (**15–21 day lead — the real clock**), fire NOC, venue/mall permissions, **music licensing (IPRS/PPL)**, noise limits.
- **First-aid / medical** — kit + a trained person; nearest hospital noted.
- **Spare-parts inventory + pit** — bulk consumables (bodyshells, wheels, arms, dogbones) so repair = swap not fix.
- **Crowd control / queue** — barriers, a heat-assignment system, wristbands, a "you're up in N heats" board.
- **Data capture** — collect emails/handles at registration; content-first = *audience-building*, so instrument it.
- **Payments** — UPI + cash handling, a POS, refund/rain policy.
- **Dead-car & jam protocols** — scripted marshal responses (§1.6).
- **Traction / dust / surface wear** — sweep the surface, manage tyre grip on astroturf.
- **Staff briefing + a full rehearsal** — run one complete mock event before doors open.
- **Accessibility** — wheelchair sightlines, seating for spectators.
- **Security** — cash, gear (cars/cameras are theft targets), overnight storage.

Applies to: [2 - TorqFall \(rulebook\)](#) · [TorqFall — Level Directory v1](#) · TorqSiege (build phase & weapons) · all future games. Feeds Event Execution Plan v1 and Build & Fabrication Plan v1.